EXP.NO: 4	System Calls
DATE:	

AIM:

To study and implement basic UNIX system calls such as process creation, execution, process identification, and directory handling using C programs.

ALGORITHM:

1. Start the program and include necessary headers (stdio.h, unistd.h, dirent.h).
2. Identify Process: Use getpid() and getppid() to fetch ID numbers from the Kernel.
3. Process Creation: Invoke fork(). Branch execution based on return value (0 for child, positive for parent).
4. Process Replacement: Use exec() family functions to replace current process image with a new executable.
5. Directory Access: Initialize a DIR pointer with opendir().
6. Read Entries: Use a while loop with readdir() to iterate through file entries until NULL is reached.
7. Cleanup & Stop: Close directory streams and terminate the program.

CODE:

Process Execution:

```
#include <stdio.h>
#include <unistd.h>

int main() {
   printf("Before exec()\n");
   execl("/bin/ls", "ls", NULL);
   printf("After exec()\n"); // This line will not execute on success
   return 0;
}
```

Process Creation:

```
#include <stdio.h>
#include <unistd.h>

int main() {
   if (fork() == 0) {
```

```
        printf("This is CHILD process\n");
    } else {
        printf("This is PARENT process\n");
    }
    printf("PID: %d\n", getpid());
    return 0;
}
```

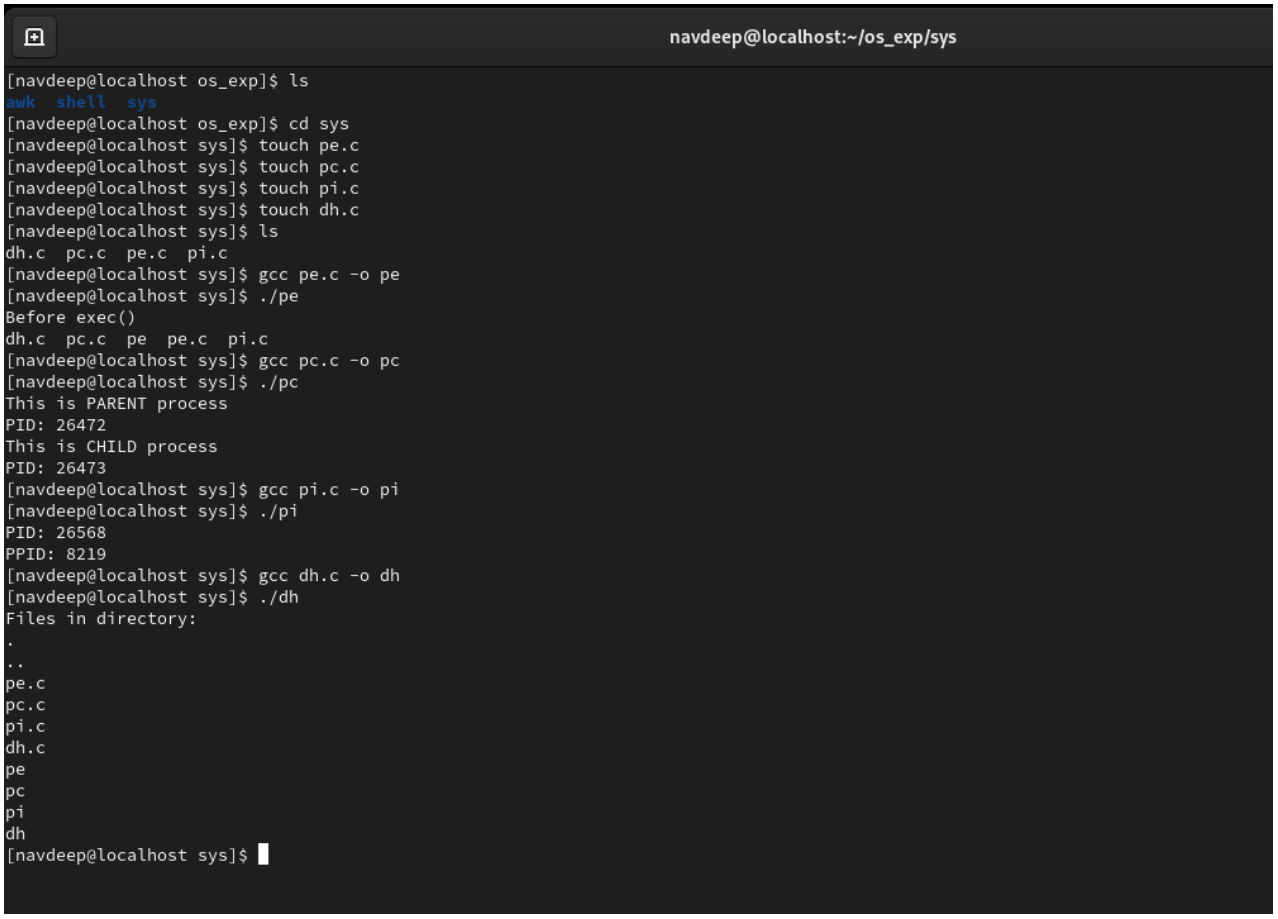
Process Identification:

```
#include <stdio.h>
#include <unistd.h>

int main() {
    printf("PID: %d\n", getpid());
    printf("PPID: %d\n", getppid());
    return 0;
}
```

Directory Handling:

```
#include <stdio.h>
#include <dirent.h>

int main() {
    DIR *dir;
    struct dirent *entry;
    dir = opendir(".");
    if (dir == NULL) {
        printf("Cannot open directory\n");
        return 1;
    }
    printf("Files in directory:\n");
    while ((entry = readdir(dir)) != NULL) {
        printf("%s\n", entry->d_name);
    }
    closedir(dir);
    return 0;
}
```

OUTPUT:



RESULT:

Thus, the basic UNIX system calls such as fork(), exec(), getpid(), and directory handling functions like opendir() and readdir() were successfully implemented and executed using C programs.